1) a)

$y^2 = 4\left(\frac{1}{3}x\right)$

$\therefore y^2 = \frac{4}{3}x$

1) b)

$(4y)^2 = 4x$

$\therefore y^2 = \frac{1}{4}x$

1) c)

$(y+1)^2 = 4(x-2)$

1) d)

$(-y)^2 = 4x$

$\therefore y^2 = 4x$

1) e)

$x^2 = 4y$

2) a)

$(2x)^2 + 3y^2 = 3$

$\therefore 4x^2 + 3y^2 = 3$

2) b)

$x^2 + 3\left(\frac{1}{5}y\right)^2 = 3$

$\therefore x^2 + \frac{3}{25}y^2 = 3$

2) c)

$(x+3)^2 + 3(y-4)^2 = 3$

2) d)

$x^2 + 3(-y)^2 = 3$

$\therefore x^2 + 3y^2 = 3$

2) e)

$y^2 + 3x^2 = 3$

3) a)

$\left(\frac{1}{3}x\right)y = 12 \therefore xy = 36$

3) b)

$x\left(\frac{1}{2}y\right) = 12 \therefore xy = 24$

3) c)

$xy = 12 \therefore (x-2)(y-2) = 12$

3) d)

$x(-y) = 12 \therefore xy = -12$

3) e)

$(-y)(-x) = 12 \therefore xy = 12$

4) a)

$\frac{(4x)^2}{4} - \frac{y^2}{5} = 1$

$\therefore 4x^2 - \frac{y^2}{5} = 1$

4) b)

$\frac{x^2}{4} - \frac{\left(\frac{1}{2}y\right)^2}{5} = 1$

$\therefore \frac{x^2}{4} - \frac{y^2}{20} = 1$

4) c)

$\frac{(x-3)^2}{4} - \frac{(y+5)^2}{5} = 1$

4) d)

$\frac{x^2}{4} - \frac{y^2}{5} = 1$

4) e)

$\frac{y^2}{4} - \frac{x^2}{5} = 1$

5) a)

$\frac{\left(\frac{1}{2}x\right)^2}{3} + \frac{y^2}{16} = 1$

$\therefore \frac{x^2}{12} + \frac{y^2}{16} = 1$, Ellipse

5) b)

$\frac{\left(\frac{1}{2}x\right)^2}{2} - \frac{y^2}{10} = 1$

$\therefore \frac{x^2}{8} - \frac{y^2}{10} = 1$, Hyperbola

5) c)

$\left(\frac{1}{2}x\right)y = 14$

$\therefore xy = 28$, Rectangular Hyperbola

5) d)

$\left(\frac{1}{2}x\right)^2 - y^2 = 4$

$\therefore \frac{x^2}{4} - y^2 = 4$, Hyperbola

6) a)

$\frac{x^2}{18} + \frac{\left(\frac{1}{3}y\right)^2}{2} = 1$

$\therefore x^2 + y^2 = 18$, Circle

6) b)

$\frac{x^2}{45} - \frac{\left(\frac{1}{3}y\right)^2}{5} = 1$

$\therefore x^2 - y^2 = 45$, Rectangular Hyperbola

6) c)

$\left(\frac{1}{3}y\right)^2 = -3x$

$\therefore y^2 = -27x$, Parabola

6) d)

$x^2 - \left(\frac{1}{3}y\right)^2 = 2$

$\therefore \frac{x^2}{2} - \frac{y^2}{18} = 1$, Hyperbola

7) a)

$\left(\frac{1}{2}x\right)^2 + y^2 = 4$

$\therefore \frac{x^2}{16} + \frac{y^2}{4} = 1$

7) b)

$x^2 + \left(\frac{1}{2}y\right)^2 = 4$

$\therefore \frac{x^2}{4} + \frac{y^2}{16} = 1$

8) a)

$x^2 = \frac{1}{2}y$

$\therefore y = 2x^2$

8) b)

$7x^2 + 3y^2 = 21$

8) c)

$xy = -6$

8) d)

$\frac{y^2}{8} - \frac{x^2}{4} = 1$

9) a)

$(-x)^2 = \frac{1}{4}(-y)$

$\therefore x^2 = -\frac{1}{4}y$

9) b)

$2(-y)^2 + 5(-x)^2 = 10$

$\therefore 2y^2 + 5x^2 = 10$

9) c)

$(-y)(-x) = \frac{1}{5}$

$\therefore xy = \frac{1}{5}$

9) d)

$\frac{(-y)^2}{3} - \frac{(-x)^2}{25} = 1$

$\therefore \frac{y^2}{3} - \frac{x^2}{25} = 1$

10) a)

Reflection in $y = x$

$\frac{x^2}{8} + \frac{y^2}{4} = 1$

10) b)

Reflection in $y = -x$

$\frac{(-y)^2}{4} + \frac{(-x)^2}{8} = 1$

$\therefore \frac{x^2}{8} + \frac{y^2}{4} = 1$

11) a)

Reflection in $y = x$

$\frac{y^2}{12} - \frac{x^2}{4} = 1$

11) b)

$x = \pm\frac{2}{\sqrt{12}}y$

So $y = \sqrt{3}x$ and $y = -\sqrt{3}x$

11) c)

A stretch, parallel to the y-axis, scale factor 2

$\frac{x^2}{12} - \frac{\left(\frac{1}{2}y\right)^2}{4} = 1 \therefore \frac{x^2}{12} - \frac{y^2}{16} = 1$

11) d)

$y = \pm\frac{4}{\sqrt{12}}x$

So $y = \frac{2\sqrt{3}}{3}x$ and $y = -\frac{2\sqrt{3}}{3}x$

12) a)

Translation $\begin{pmatrix}0\\3\end{pmatrix}$

12) b)

Translation $\begin{pmatrix}0\\-1\end{pmatrix}$

12) c)

Translation $\begin{pmatrix}1\\-3\end{pmatrix}$

12) d)

A stretch, parallel to the x-axis, scale factor $\frac{1}{2}$

12) e)

A stretch, parallel to the x-axis, scale factor 3

12) f)

$\frac{x^2}{16} - (y+2)^2 = 16$

Translation $\begin{pmatrix} 0 \\ -2 \end{pmatrix}$

12) g)

$\frac{(x+5)^2}{8} - \frac{(y-1)^2}{2} = 1$

Translation $\begin{pmatrix} -5 \\ 1 \end{pmatrix}$

12) h)

$\frac{x^2}{16} - 18y^2 = 1$

A stretch, parallel to the y-axis, scale factor $\frac{1}{6}$

12) i)

$2x^2 - 12x - 3y^2 - 18y = 21$

$2(x^2 - 6x) - 3(y^2 + 6y) = 21$

$2((x-3)^2 - 9) - 3((y+3)^2 - 9) = 21$

$2(x-3)^2 - 18 - 3(y+3)^2 + 27 = 21$

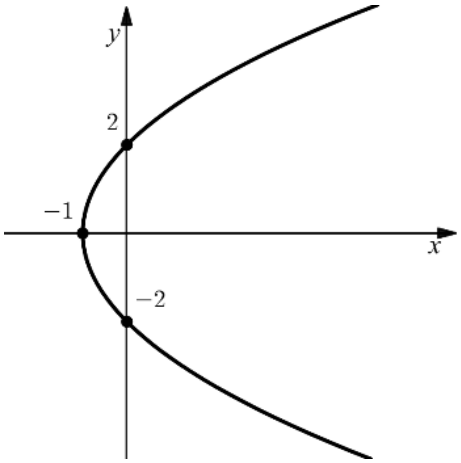
$2(x-3)^2 - 3(y+3)^2 = 12$

$\frac{(x-3)^2}{6} - \frac{(y+3)^2}{4} = 1$

Translation $\begin{pmatrix} 3 \\ -3 \end{pmatrix}$

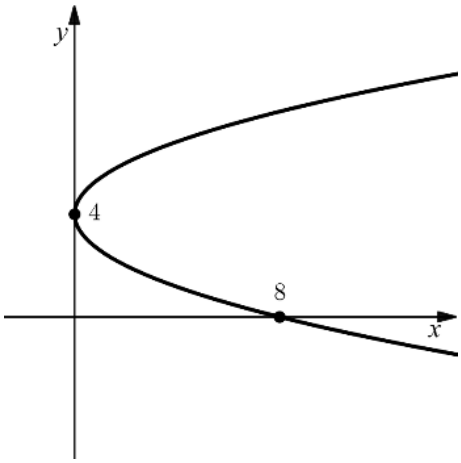
13) a)

When $x = 0$, $y = \pm 2$, when $y = 0$, $x = -1$



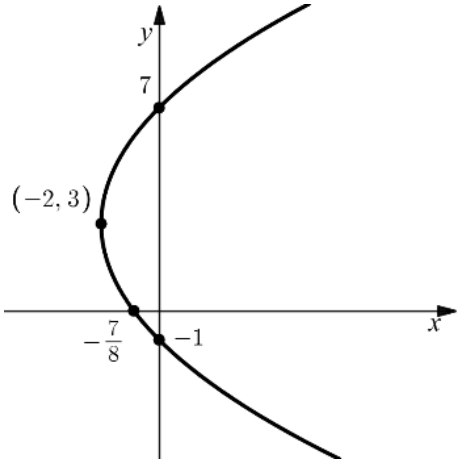
13) b)

When $x = 0$, $y = 4$, when $y = 0$, $x = -1$



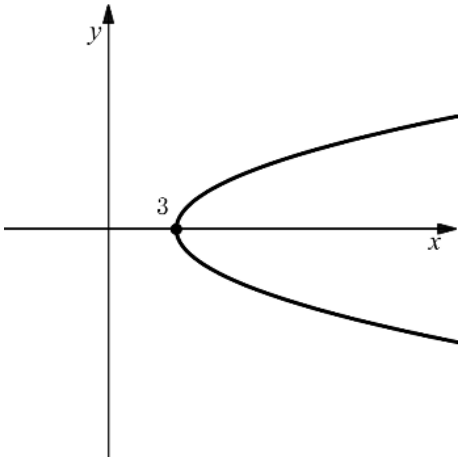
13) c)

When $x = 0$, $y = -1$ or $y = 7$, when $y = 0$, $x = -\frac{7}{8}$



13) d)

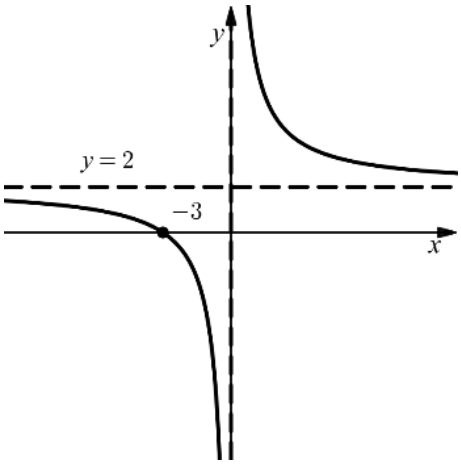
When $x = 0$, no solutions when $y = 0$, $x = 3$



14) a)

When $x = 0$, no solutions, when $y = 0$, $x = -3$

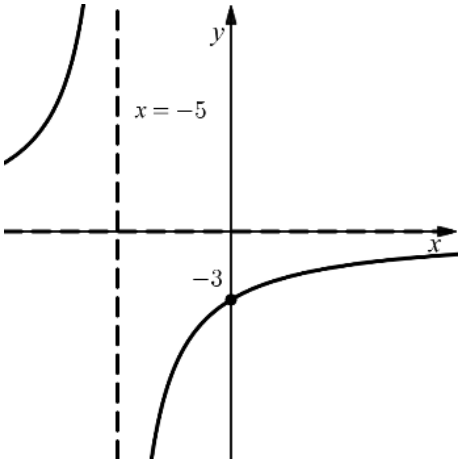
Asymptotes: $x = 0$, $y = 2$



14) b)

When $x = 0$, $y = -3$, when $y = 0$, no solutions

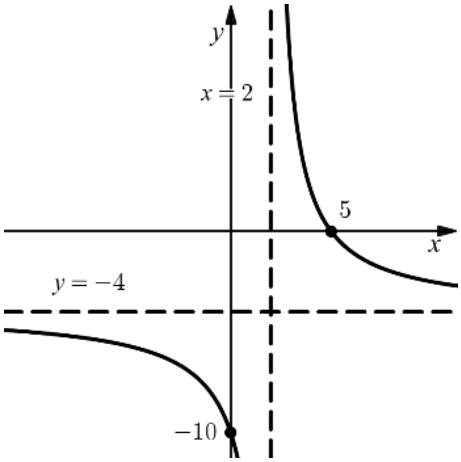
Asymptotes: $x = -5$, $y = 0$



14) c)

When $x = 0$, $y = -10$, When $y = 0$, $x = 5$

Asymptotes: $x = 2$, $y = -4$

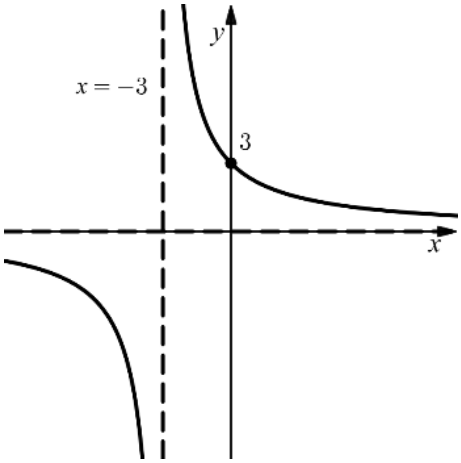


14) d)

$y(3+x)=9$

When $x=0$, $y=3$, when $y=0$, no solutions

Asymptotes: $x=-3$, $y=0$

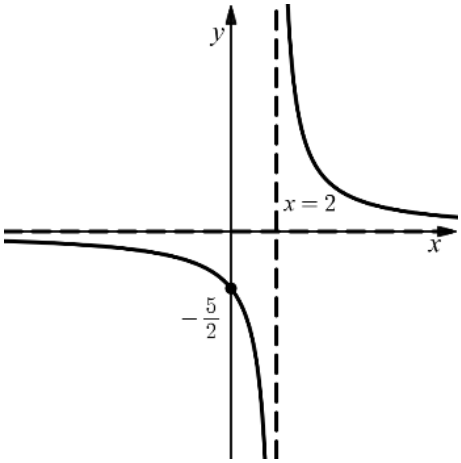


14) e)

$y(x-2)=5$

When $x=0$, $y=-\frac{5}{2}$, when $y=0$, no solutions

Asymptotes: $x=-3$, $y=0$



15) a)

$\frac{x^2}{25}+\frac{(y+1)^2}{5}=1$

Translation $\begin{pmatrix}0\\-1\end{pmatrix}$

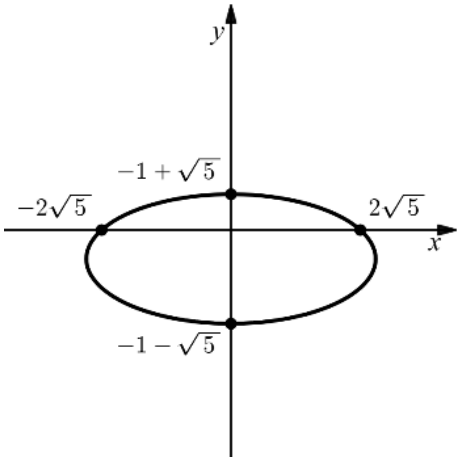
15) b)

When $x = 0$, $(y+1)^2 = 5$ so $y = -1 \pm \sqrt{5}$,

When $y = 0$, $x^2 = 20$ so $x = \pm 2\sqrt{5}$

$(0, -1-\sqrt{5})$, $(0, -1+\sqrt{5})$, $(2\sqrt{5}, 0)$ and $(-2\sqrt{5}, 0)$

15) c)



16) a)

Hyperbola

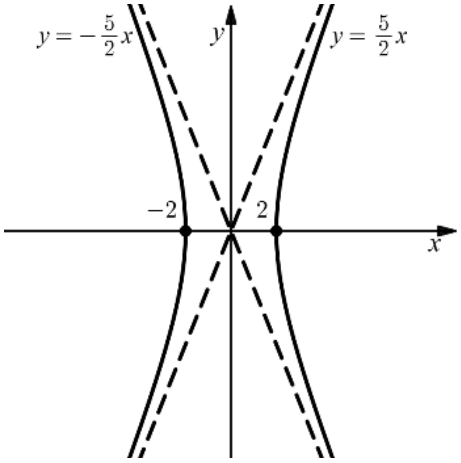
16) b)

$y = \frac{2\sqrt{3}}{3}x$ and $y = -\frac{2\sqrt{3}}{3}x$

16) c)

$y = \frac{2\sqrt{3}}{3}(x-4)$ and $y = -\frac{2\sqrt{3}}{3}(x-4)$

17) a)



17) b)

$k = 2$

17) c)

$y = \frac{5}{2}x$ and $y = -\frac{5}{2}x$ map to $y = \frac{5}{2}(x-2)$ and $y = -\frac{5}{2}(x-2)$

$y = \frac{5}{2}x - 5$ and $y = -\frac{5}{2}x + 5$

18) a)

When $x = 0$, $y = \pm 1$, when $y = 0$, $x = \pm 3$

$(0, -1)$, $(0, 1)$, $(3, 0)$ and $(-3, 0)$

18) b)

$x^2 - 4x + 9y^2 = 5$

$(x-2)^2 - 4 + 9y^2 = 5$

$(x-2)^2 + 9y^2 = 9$

$\frac{(x-2)^2}{9} + y^2 = 1$

Translation $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$

18) c)

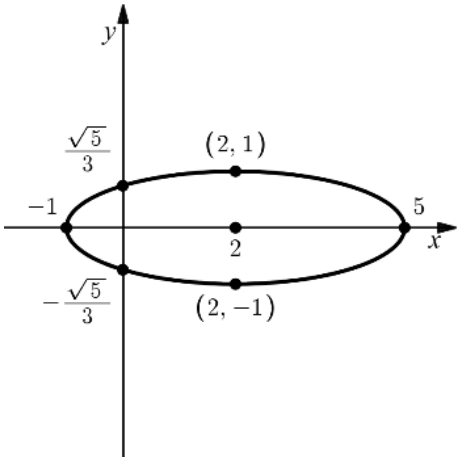
$(2, -1)$, $(2, 1)$, $(5, 0)$ and $(-1, 0)$

18) d)

When $x = 0$, $y = \pm\frac{\sqrt{5}}{3}$, when $y = 0$, $x = -1$, $x = 5$

$\left(0, -\frac{\sqrt{5}}{3}\right)$, $\left(0, \frac{\sqrt{5}}{3}\right)$, $(5, 0)$ and $(-1, 0)$

18) e)



19) a)

$y = \frac{2}{3}x$ and $y = -\frac{2}{3}x$

19) b)

Translation $\begin{pmatrix} -4 \\ 0 \end{pmatrix}$

19) c)

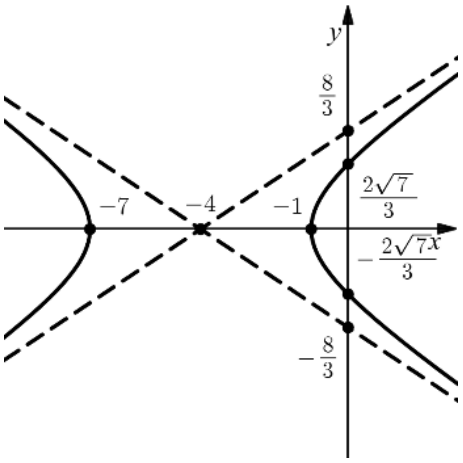
$y = \frac{2}{3}(x + 4)$ and $y = -\frac{2}{3}(x + 4)$

19) d)

When $x = 0$, $\frac{16}{9} - \frac{y^2}{4} = 1 \therefore y = \pm\frac{2\sqrt{7}}{3}$, when $y = 0$, $(x + 4)^2 = 9 \therefore x = -1$ or $x = -7$

$\left(0, \frac{2\sqrt{7}}{3}\right)$, $\left(0, -\frac{2\sqrt{7}}{3}\right)$, $(-1,0)$ and $(-7, 0)$

19) e)



20) a)

$y - 2 = \frac{5}{2}x$ and $y - 2 = -\frac{5}{2}x$

So $y = \frac{5}{2}x + 2$ and $y = -\frac{5}{2}x + 2$

20) b)

Translation $\begin{pmatrix} 0 \\ -5 \end{pmatrix}$

20) c)

$y + 5 = \frac{5}{2}x + 2$ and $y + 5 = -\frac{5}{2}x + 2$

So $y = \frac{5}{2}x - 3$ and $y = -\frac{5}{2}x - 3$

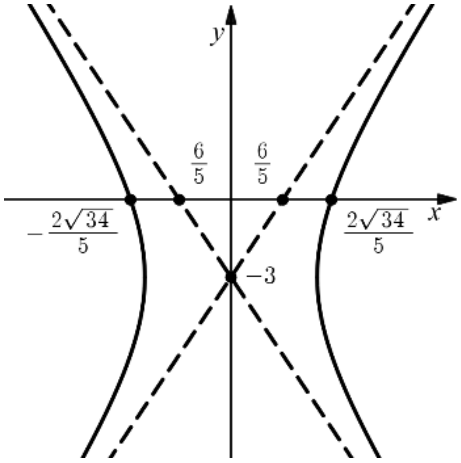
20) d)

When $x = 0$, $(y + 3)^2 = -25$ so no solutions

When $y = 0$, $\frac{x^2}{4} - \frac{9}{25} = 1 \therefore x = \pm\frac{2\sqrt{34}}{5}$

$\left(\frac{2\sqrt{34}}{5}, 0\right)$ and $\left(-\frac{2\sqrt{34}}{5}, 0\right)$

20) e)



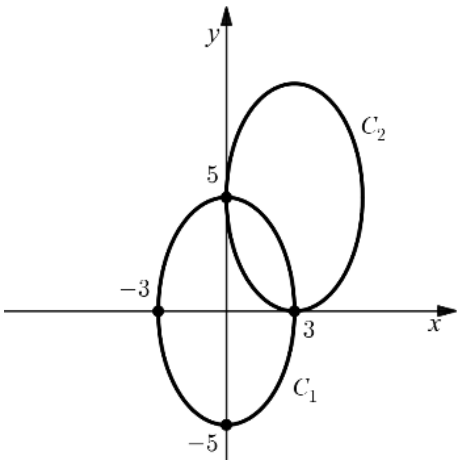
21) a)

$\begin{pmatrix}3\\5\end{pmatrix}$

21) b)

$(3,0)$ and $(0,5)$

21) c)



21) d)

The tangent lines have gradient $\frac{5}{3}$

21) e)

For $y = \frac{5}{3}x + k$

With C_1

$\frac{x^2}{9} + \frac{\left(\frac{5}{3}x+k\right)^2}{25} = 1 \Rightarrow 25x^2 + 9\left(\frac{5}{3}x + k\right)^2 = 225$

$\Rightarrow 25x^2 + 9\left(\frac{25}{9}x^2 + \frac{10}{3}kx + k^2\right) = 225$

$\Rightarrow 25x^2 + 25x^2 + 30kx + 9k^2 = 225$

$\Rightarrow 50x^2 + 30kx + 9k^2 - 225 = 0$

Discriminant: $(30k)^2 - 4 \times 50(9k^2 - 225) = 0$

$-900k^2 + 45000 = 0$

$\therefore k = \pm\sqrt{50} = \pm 5\sqrt{2}$

So the tangent lines are $y = \frac{5}{3}x + 5\sqrt{2}$ and $y = \frac{5}{3}x - 5\sqrt{2}$

22) a)

When $x = 0$, $y = \pm\sqrt{3}$ and when $y = 0$ $x = \pm 2\sqrt{2}$

$(0, -\sqrt{3})$, $(0, \sqrt{3})$, $(2\sqrt{2}, 0)$ and $(-2\sqrt{2}, 0)$

22) b)

$\frac{(x-k)^2}{8} + \frac{y^2}{3} = 1$

22) c)

$\frac{(x-k)^2}{8} + \frac{(2x+1)^2}{3} = 1$

$3(x-k)^2 + 8(2x+1)^2 = 24$

$3(x^2 - 2kx + k^2) + 8(4x^2 + 4x + 1) = 24$

$3x^2 - 6kx + 3k^2 + 32x^2 + 32x + 8 = 24$

$35x^2 + (32 - 6k)x + (3k^2 - 16) = 0$ as req.

22) d)

Tangent $\therefore b^2 - 4ac = 0$

$(32 - 6k)^2 - 4 \times 35 \times (3k^2 - 16) = 0$

$36k^2 - 384k + 1024 - 420k^2 + 2240 = 0$

$-384k^2 - 384k + 3264 = 0$

$2k^2 + 2k - 17 = 0$

$k = -\frac{1}{2} \pm \frac{\sqrt{35}}{2}$

So since k is positive $k = -\frac{1}{2} + \frac{\sqrt{35}}{2}$

23) a)

$\left(\frac{1}{3}x\right)^2 + y^2 = 10$

$\therefore \frac{x^2}{9} + y^2 = 10$

23) b)

$3\left(\frac{1}{3}x\right) - y = 10$

$\therefore x - y = 10$

23) c)

$\frac{x^2}{9} + (x - 10)^2 = 10$

$\frac{x^2}{9} + x^2 - 20x + 100 = 10$

$x^2 + 9x^2 - 180x + 900 = 90$

$x^2 - 18x + 81 = 0$

$b^2 - 4ac = (-18)^2 - 4 \times 1 \times 81 = 324 - 324 = 0$

One solution, so the line is a tangent to the curve

They intersect when $x = 9$, $y = -1$

$(9, -1)$

23) d)

$\frac{x^2}{90} + \frac{y^2}{10} = 1$ and $x - y = 10$

